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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This supplement retains optimal-advantage centering and every preceding complete proof, including constructive qualification, original investment actors and original-law inference. It adds proofs of complete-policy occupation transfer, nonlinear recursive comparison, strategic deviation certificates, joint-domain neural construction, adaptive acquisition coverage, predictor work bounds, dated accuracy allocation, prospective menu regret and uniform price-fee sensitivity. Counterexamples isolate incorrect reference-policy weighting and unattainable shared-readout combinations. Historical numerical evidence remains in the Numerical and Development Appendix. It additionally supplies the full pattern-disaggregation, coupled-successor-cost and original-law transport-qualification proofs, including the stationary discounted extension. The signed recursive qualification theorem, sharpness examples and risk-specific transport proofs are included in full.

1. POLICY-COST INFERENCE AT THE SCALE OF VERIFIED REGRET

The query certificate and a direct cost experiment answer different questions. Nevertheless, a certificate can improve the experiment without replacing its estimand. The relevant observation is a sum of optimal Bellman advantages along the actual policy trajectory. Its expectation is policy loss, while its range is controlled by the local accuracy contract rather than by the range of economic costs. All conditional expectations used to construct this observation must themselves be evaluated and charged.

Write V_t for the original Bellman optimum, $V_T = g$, $b_0 = 1$, and $b_t = \prod_{j<t} \beta_j$. For a feasible frozen policy π, define

$$a_t^\pi(x) = c_t(x, \pi_t(x)) + \beta_t P_t^{\pi_t(x)} V_{t+1}(x) - V_t(x), \qquad Z^\pi = \sum_{t<T} b_t a_t^\pi(X_t^\pi).$$

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The symbol a_t^π denotes an advantage, not an action. Assume a verified local bound $0 \le a_t^\pi(x) \le e_t^\pi$ for every true state, including the acquisition allowance. Put $G_\pi = \sum_{t<T} b_t e_t^\pi$. A sample contains the actual trajectory of each compared controller under the original law; common initial states and common innovations may couple the controllers. Different sample rows are independent conditional on the frozen construction.

THEOREM 1—Bellman-centered cost comparison: *For two such controllers π, ν with a common initial law λ, let $D = Z^\pi - Z^\nu$. Then*

$$\mathbb{E}D = \mathbb{E}_\lambda[J_0^\pi - J_0^\nu], \qquad -G_\nu \le D \le G_\pi. \tag{1}$$

Suppose deterministic computation encloses D_i by $[L_i, U_i] \subseteq [-G_\nu, G_\pi]$ on every sampled continuous bin, including every compatible acquired action. For a conditionally fixed finite family of M contrasts, put $R = G_\pi + G_\nu$ and

$$h_n = R\sqrt{\frac{\log(2M/\alpha)}{2n}}.$$

With probability at least $1-\alpha$, simultaneously for the family,

$$\mathbb{E}_\lambda[J_0^\pi - J_0^\nu] \in \left[\frac{1}{n}\sum_i L_i - h_n, \frac{1}{n}\sum_i U_i + h_n\right]. \tag{2}$$

The interval may be intersected with $[-G_\nu, G_\pi]$. Its width is the mean numerical enclosure width plus $2h_n$. More generally, any valid endpoint-sample concentration bound may replace the bounded-sum radius.

Proof. See Supplement, Section 3.5.

The construction is related to martingale control variates and sequential value-based evaluation, including Jiang and Li (2016). It is not an off-policy importance-weighting argument and does not assume that a fitted value is correct. The additional requirements here are an original-optimum advantage enclosure, a true-state local regret bound, and a paid numerical evaluation of the conditional value. The target in (2) is the original expected cost difference, not the difference between

two regret certificates. The realized centered observation generally differs from the realized cash-cost difference; their expectations agree by the displayed identity.

1.1. *Numerical construction and a target-precision rule*

The returned action is fixed before evaluation. At a center $\bar{x}$ of a true-state box, let $[Q_L, Q_U]$ enclose $Q_t^*(\bar{x}, \pi_t(x))$ for that fixed action and let $[V_L, V_U]$ enclose $V_t(\bar{x})$. If L_Q, L_V are state Lipschitz bounds and ρ bounds $\|x - \bar{x}\|$, then

$$[Q_L - V_U - (L_Q + L_V)\rho, Q_U - V_L + (L_Q + L_V)\rho] \cap [0, e_t^\pi] \tag{3}$$

is a valid advantage enclosure. An empty intersection stops verification. Actions must be feasible at every state used by the endpoint queries. If the box crosses an acquisition boundary, one may enumerate all compatible acquired actions, or use $[0, e_t^\pi]$ for the unresolved part. The latter is conservative and its width remains in the reported numerical account. It is not permissible to discard that trajectory.

For the scalar investment economy, at r remaining dates we use

$$L_Q = \frac{8}{d} + \frac{11\beta}{16} G_{r-1}, \qquad L_V = G_r, \qquad e_t^\pi = \delta_t + d(L_Q + L_V)2^{-b},$$

from the original acquisition argument. The evaluator queries the optimum with its own, smaller tolerance; it does not retrain the policy or change the action being evaluated. Original-law quadrature, action-cover error and arithmetic are already included in $[Q_L, Q_U]$ and $[V_L, V_U]$. They must not be omitted or counted again as independent random noise.

COROLLARY 2—Precision and noninferiority: *Fix a numerical half-width allocation $v > 0$ and a sampling allocation $s > 0$ before observing the new contrasts. If $n^{-1}\sum_i (U_i - L_i) \le 2v$ and*

$$n \ge \frac{R^2 \log(2M/\alpha)}{2s^2},$$

then the interval in (2) has half-width at most $v + s$. For a prespecified loss margin $m > 0$, an upper endpoint at most m establishes noninferiority at that margin; containment in $[-m, m]$ establishes the corresponding two-sided equivalence statement on the same event. Neither statement identifies a strict advantage.

Proof. See Supplement, Section 3.6.

Both n and evaluator precision have costs. The deposited design fixes them prospectively and reports failure to meet the width allocation rather than increasing the sample after inspecting a favorable sign. The actual numerical width is decomposed into the conditional Bellman enclosure, the state-bin Lipschitz expansion, unresolved acquisition branches and endpoint arithmetic. The conditional Bellman component includes its action-cover and continuous-shock quadrature work. A smaller statistical support is therefore not advertised as a free runtime improvement.

1.2. *The adoption decision after inference*

Let $[l, u]$ enclose $\mathbb{E}[J^\pi - J^\nu]$, let W_π, W_ν be measured complete deployment work under a declared boundary, and let $p \geq 0$ be a decision maker's price per unit of that work. Replacing ν by π at fee $f \geq 0$ has net-cost difference in

$$[l + p(W_\pi - W_\nu) + f,\ u + p(W_\pi - W_\nu) + f].$$

A negative upper endpoint licenses the replacement under those declared prices. A noninferiority finding permits a separate assessment of computational savings; it does not convert a normalized loss margin into an estimated monetary value. This keeps the original economic purchase question while making its statistical precision and its required price inputs explicit.

2. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

2.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

2.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

2.3. *Proof of the result 3*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

2.4. *Proof of the result 4*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta)w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x - y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right

side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (10). *Q.E.D.*

2.5. *Proof of the result 5*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\hat{\pi}}$ is used. *Q.E.D.*

2.6. *Proof of the result 13*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

2.7. *Proof of the result 14*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center

value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$ cannot be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (21) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (22). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

2.8. *Proof of the result 4*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem 14 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and

newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

2.9. *Proof of Proposition 5*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (7). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear

term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (8). *Q.E.D.*

2.10. *Proof of the result 1*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

2.11. *Proof of the result 2*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (3). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

2.12. *Proof of the result 3*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

3. PROOFS OF THE OPERATIONAL AND NEW RESULTS

3.1. *Proof of the result 15*

PROOF: Strong convexity gives $f((1-s)l+su) \le (1-s)f(l)+sf(u)-\mu s(1-s)(u-l)^2/2$. Replacing the endpoint values by their upper bounds proves (24). Subtraction of a valid lower bound proves (25). A failed screen adds no endpoint to the partition and alters no lower bound. Subsequent protected refinement is therefore covered by the earlier termination proof. A successful screen supplies exactly the local accuracy premise used in the acquired-policy recursion. Finally, C is a convex quadratic in a, proving the sublevel-set assertion. *Q.E.D.*

3.2. *Proof of the result 16*

PROOF: A successful screen returns without an endpoint query. A failed screen invokes a procedure bounded by $K(z)$, and its margin loss equals one. This proves (28). For each fixed predictor the variable $K(z)\ell_\tau(z,\theta)$ lies in $[0, K_{\max}]$. The bounded-sum inequality of Hoeffding (1963) and a union bound over the conditionally fixed family give (29). *Q.E.D.*

3.3. *Proof of the result 17*

PROOF: The chord is a convex quadratic in its normalized action, which is affine in w. Squaring the positive part preserves convexity and gives a continuously differentiable function. The nonnegative weighted sum and quadratic penalty preserve these properties. Convexity gives $F(w)-F(v) \le g^\top(w-v)$, and the minimum of $g^\top v$ on the box is $\sum_j \min\{0, g_j\}$. This proves (31). Every failed screen has squared positive-part residual at least τ^2, while every successful screen uses zero further endpoint queries. Summing the corresponding K_i bounds proves the work claim. Strong convexity with modulus λ proves the distance bound. *Q.E.D.*

3.4. *Proof of the result 18*

PROOF: Semiconcavity means that $f(a) - H\|a\|_2^2/2$ is concave. Its barycentric inequality, with vertex values replaced by lower endpoints, gives (32). The tangent inequality for the convex quadratic p proves (33); a linear function attains its simplex

minimum at a vertex. Taking minima over a cover preserves the lower bound. Each feasible upper witness is an action of the actual model. The local gap premise therefore implies the earlier policy recursion, irrespective of how the triangle or support point was proposed.

For finiteness, use simplex lattice denominator A, with $(A+1)(A+2)/2$ vertices. On triangles of diameter proportional to $c(x)/A$, the curvature error tends to zero as A^{-2}. The endpoint error is reduced by finite descendant accuracy and quadrature refinements. The retained exact-rational recovery contract requires $\delta > 16\Lambda\, 2^{-32}$ at each recursive call, leaving positive slack after rounding; dyadic enlargement of A and q then terminates. This is a finite arithmetic procedure, not a uniform bound on rational bit cost. For (34), the independent innovation variances after conditioning on the bins are $\rho_j^2/(3q^2)$, their direction norms squared are d, and the semiconcavity Taylor remainder is $M_{t+1}/2$ times their sum. Substitution of $\rho_1 = 1/32$, $\rho_2 = 1/64$ gives the coefficient. Finally, each action query has q_r^2 descendants, proving (35). *Q.E.D.*

3.5. *Proof of the result 1*

PROOF: Conditional expectation and $b_{t+1} = b_t\beta_t$ imply

$$\mathbb{E}\sum_{t<T} b_t a_t^{\pi}(X_t^{\pi}) = \mathbb{E}\left[\sum_{t<T} b_t c_t(X_t^{\pi}, \pi_t(X_t^{\pi})) + b_T g(X_T^{\pi}) - V_0(X_0^{\pi})\right].$$

The intermediate value terms cancel. Subtracting the two expectations cancels the initial value under the common initial law, proving the identity. Summing the pointwise local bounds proves the range. Applying the bounded-sum inequality to each lower and upper endpoint sample, with failure allocation $\alpha/(2M)$, and taking a union bound proves (2). Endpoint containment supplies the comparison between endpoint expectations and the target. The width identity follows by subtraction. No pathwise identity between D and the realized cumulative-cost difference is asserted. *Q.E.D.*

3.6. *Proof of the result 2*

PROOF: The sample-size condition implies $h_n \le s$. Insert this and the numerical-width condition into the width identity. The final claims are set inclusions for the simultaneous confidence interval. *Q.E.D.*

3.7. *Proof of the result 8*

PROOF: The bounded-sum argument in Theorem 16 gives $Pf_\theta \le u_\theta$ for every θ on one event of probability at least $1-\alpha$. For $0 \le f \le K_{\max}$, the layer-cake representation gives $|Pf - Qf| \le K_{\max}\|P - Q\|_{\mathrm{TV}}$. Hence, on that event,

$$\mathbb{E}[N_{\theta_b}(Z_b) \mid \mathcal{F}_{b-1}] \le \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\}.$$

The variables $c_b N_{\theta_b}(Z_b)$ have conditional ranges of length at most $c_b K_{\max}$. Conditional Hoeffding bounds, multiplied successively along the filtration and followed by the exponential Markov inequality, show that the sum exceeds its conditional means by at most the final term of (11), except on an event of probability γ. Union with the validation failure event proves the result. Every selected readout remains a proposal to the same verified gate. Its statewise certificate, and thus the earlier acquired-policy recursion, is unchanged. *Q.E.D.*

3.8. *Proof of the result 9*

PROOF: The expected cost difference in a normal block is $m\Delta$; it is zero in a declined block. Linearity of expectation gives $m(B-D)\Delta$, without requiring block independence. Multiply the cost interval by its nonnegative price, and add the work difference and fee. All price choices use the same event, so no additional probability allocation is needed. *Q.E.D.*

4. RETAINED INTERVENING TRAINING RESULTS

4.1. *Proof of Proposition 6*

PROOF: A successful screen incurs no recovery queries. On a failed screen the clipped loss equals one, and recovery work is at most WK. The pointwise bound therefore holds for the bounded variable $WK\ell_\tau$. The bounded-sum inequality and a

union bound give the claim. None of the separately named service costs is included in the definition of recovery work. *Q.E.D.*

4.2. *Proof of Proposition 7*

PROOF: The lower cover still covers the entire feasible set, so its unchanged lower bound remains valid. Each additional upper endpoint bounds the true cost of its own feasible action, hence is an upper bound for the infimum. Their minimum preserves this property. The returned action's true cost is no greater than its own upper endpoint, proving the regret claim. *Q.E.D.*

5. PROOFS FOR FROZEN CONTINUATIONS AND PURCHASED WITNESSES

5.1. *Frozen-continuation comparison*

PROOF OF THEOREM 6: Fix the construction information. Conditional expectation at date t gives

$$\mathbb{E}[b_t \beta_t P_t^{\pi_t} h_{t+1}(X_t)] = \mathbb{E}[b_{t+1} h_{t+1}(X_{t+1})].$$

Summing these equalities cancels all intermediate continuation terms in (14); the terminal correction gives

$$\mathbb{E} Z_h^{\pi} = \mathbb{E} J_0^{\pi} - \mathbb{E} h_0(X_0^{\pi}).$$

The common initial law cancels the last expectation when policies are compared. Their trajectory coupling otherwise does not matter for this identity.

Pointwise, $Q_t^h(x, \pi_t(x)) - h_t(x) = r_t^h(x) + a_t^{h,\pi}(x)$. For any two trajectory realizations, the difference of their residual terms is between $-\omega_t$ and ω_t. Their greedy-gap difference is between $-\eta_t^{\nu}$ and η_t^{π}. The same oscillation argument applies to $g - h_T$ at the terminal states. Multiplication by the nonnegative discounts and summation proves (16).

For the regret assertion, choose constants $\underline{r}_t, \overline{r}_t$ enclosing r_t^h with difference at most ω_t; an arbitrarily small enlargement handles unattained infima and suprema. Likewise enclose the terminal correction by $\underline{z}, \overline{z}$. For every feasible policy μ, $a_t^{h,\mu} \geq 0$.

Telescoping therefore gives, at each initial state,

$$J_0^{\mu} \geq h_0 + \sum_{t<T} b_t \underline{r}_t + b_T \underline{z}, \qquad J_0^{\pi} \leq h_0 + \sum_{t<T} b_t (\bar{r}_t + \eta_t^{\pi}) + b_T \bar{z}.$$

Taking the infimum over μ in the first inequality, subtracting it from the second, and then removing the enlargement proves (17). No minimizing action or optimal-policy attainment is needed. If h_t is replaced by $h_t + k_t$, then $Q_t^h - h_t$ increases by $\beta_t k_{t+1} - k_t$, and the terminal correction decreases by k_T. Their discounted sum changes by $-k_0$, identically on both trajectories. Residual oscillations and greedy gaps are unchanged. With $h = V^*$ all residual and terminal oscillations vanish. *Q.E.D.*

For completeness, the confidence statement is conditional on both policy and continuation construction. After intersection with the true support, each endpoint sample lies in an interval of length R_h. A lower-tail or upper-tail bounded-sum bound with failure allocation $\alpha/(2M)$ gives the two ends of (18). Endpoint containment implies $\mathbb{E}L_i \leq \mathbb{E}D_h \leq \mathbb{E}U_i$. A union bound proves simultaneous coverage. Dependence within a paired trajectory is permitted; independence is required across the sampled rows under this construction. A predictor or cache modified in response to earlier inference rows is not conditionally fixed by the original training information.

5.2. *Exact conditional moments and uniform qualification*

PROOF OF PROPOSITION 7: If U is uniform on $[-A, A]$, integration of $(z+u)_+$ gives

$$\frac{1}{2A} \int_{-A}^{A} (z+u)_+ \, du = \frac{(z+A)_+^2 - (z-A)_+^2}{4A}.$$

A second integration with respect to an independent uniform variable on $[-B, B]$ integrates each positive-part square to a positive-part cube divided by three. Dividing by both uniform densities gives (20). Vanishing radii correspond to deterministic projections and must be removed before division. Symmetry of the innovation permits each nonzero projection coefficient to be replaced by its absolute value.

In a rational shallow network every projected center and radius is rational. Positive parts, powers, sums and divisions by checked nonzero rationals are exact

rational operations. The affine part has its usual conditional mean, and linearity of integration allows signed readout coefficients. Computing all K projections at a state and action takes $O(Kd)$ arithmetic operations; each subsequent one- or two-shock neuron integral takes a fixed number of operations. Summation along n trajectories of length T gives the stated count. This says nothing about the size of the numerators and denominators or about the work used to construct and qualify h. *Q.E.D.*

Proof of the residual qualification account. Let $v_t^h(x) = \inf_{a \in A(x)} Q_t^h(x,a)$. Given an action $a \in A(y)$, proportional reduction to $A(x)$ has ℓ^1 displacement $(\sum_i a_i - c(x))_+$, at most $(c(y) - c(x))_+ \le \kappa \|x - y\|_1$. Hence

$$v_t^h(x) \le Q_t^h(y,a) + (L_{x,t} + \kappa L_{a,t}) \|x - y\|_1.$$

Taking the infimum over a, and reversing x, y, proves the asserted modulus of v_t^h. Subtracting h_t proves that of r_t^h. Every point of a radius-ρ state cover is within ρ of a cover point, so its endpoint expansion yields (21). For an action mesh with fill radius s, approximate any feasible action by a feasible mesh action within s. The Lipschitz inequality gives $\min_{\text{mesh}} Q_t^h - L_{a,t} s \le v_t^h \le \min_{\text{mesh}} Q_t^h$. Outward action endpoints preserve both inequalities. There are $N_t A_t$ such conditional evaluations per date. Actor-specific greedy gaps and the terminal correction are additional certificates, not consequences of checking only the residual at the cover points.

5.3. *Transport and actual-policy validity*

PROOF OF THEOREM 10: The value modulus gives $V_r^*(x) \ge L_y - L_V D$. Nonnegative economic costs give the zero lower bound. Variation first in state and then in action yields

$$Q_r^*(x, \widetilde{a}) \le Q_r^*(y, a_y) + L_Q D + L_A \|\widetilde{a} - a_y\|_1 \le U(x).$$

Both state endpoints and the repaired action lie in the ambient state–action domain on which these moduli are valid. Feasibility of $\widetilde{a}$ supplies $V_r^*(x) \le Q_r^*(x, \widetilde{a})$. Outward arithmetic preserves the two end inequalities; testing the resulting width, rather than an inward-rounded width, gives the local gap.

At each history and date, the returned action has original-optimum advantage no greater than its certified local gap and the stated acquisition allowance. Conditional expectation of the next value therefore gives the same one-step regret recursion used in Theorem 14. Backward induction proves the full discounted loss sum, even when the action selected at x depends on which valid records earlier requests purchased. No equality with the fresh-search action is used. In particular, a population policy-cost interval for a different, frozen controller cannot be borrowed merely because both controllers have the same local-gap upper bound. *Q.E.D.*

The moduli can also be checked directly from the inherited primitive proof. The state cost has ℓ^1 Lipschitz constant $8/d$; the transition's induced ℓ^1 state norm is at most $11/16$. The action cost on the ambient nonnegative simplex of total mass at most $1/4$ has component derivatives bounded by $13/16$, and each exposure column has ℓ^1 norm $3d/8$. Composition with the G_{r-1}-Lipschitz true continuation gives L_Q, L_A. The value modulus G_r is Theorem 4. These constants do not depend on a fitted predictor.

5.4. *Capacity rounding and the fill bound*

PROOF OF PROPOSITION 11: Let $c_q(x) = q\lfloor c(x)/q \rfloor$. If an already lattice-aligned a_y has total mass no greater than $c_q(x)$, it is unchanged. Otherwise proportional reduction removes total mass $\sum_i (a_y)_i - c_q(x)$, and downward rounding removes less than mq more. Since $c_q(x) > c(x) - q$ and $a_y \in A(y)$, the total displacement is at most $\kappa D + (m+1)q$.

Before the favorable zero truncation of the lower bound, the expanded width is no greater than

$$\gamma\delta + (L_V + L_Q + \kappa L_A)D + L_A(m+1)q + \xi.$$

It is at most δ for $D \le \rho_*$. Because all records are checked and retained, two centers at which fresh fills are purchased must consequently be more than ρ_* apart in ℓ^1. Partition each coordinate of the unit cube into $n = \lceil d/\rho_* \rceil$ equal intervals, with a fixed boundary convention. The diameter of each resulting box is at most $d/n \le \rho_*$. It can contain at most one fresh center. Thus $M_r \le n^d \le (1 + d/\rho_*)^d$. Each fill is

paid once, whereas every query incurs its actual lookup, verification and record handling. Adding these operations proves (16). If records are evicted or invalidated, the separation argument applies only within a period in which the required earlier records remain available; lost coverage must be repurchased and charged. *Q.E.D.*

6. PROOFS FOR QUALIFICATION-OPTIMAL READOUTS AND SEQUENTIAL COSTS

6.1. *The readout program and its dual*

PROOF OF THEOREM 8: The readout has Lipschitz constant at most $L(\theta) = \sum_k L_k|\theta_k|$. At a cover point the residual is in $[y_j^- - F_j\theta, y_j^+ - F_j\theta]$. The Lipschitz constant of that residual is at most $C + L(\theta)$. For any state choose a cover point at distance at most ρ. Its lower endpoint minus $\rho(C + L(\theta))$ is a valid lower bound there, and the upper endpoint plus that quantity is a valid upper bound. Taking the minimum and maximum over the cover gives the asserted oscillation bound. No distribution of states, interpolation of a discontinuous actor, or convexity of the fitted continuation is used.

Introduce $u, l \in \mathbb{R}$ and $s_k \geq 0$ and minimize

$$u - l + 2\rho C + \sum_k \lambda_k s_k$$

subject to

$$\begin{aligned} &u \geq y_j^+ - F_j\theta, && l \leq y_j^- - F_j\theta && (j = 1, \ldots, N), \\ &s_k \geq \theta_k, && s_k \geq -\theta_k && (k = 1, \ldots, K), \end{aligned}$$

and $-B_k \leq \theta_k \leq B_k$. The stated dimensions follow. For any θ the tight choices of u, l and the choice $s_k = |\theta_k|$ attain (22); conversely every feasible point has objective at least that expression. Continuity on the compact coefficient box gives attainment.

The maximum of finitely many real numbers is the maximum of their averages over probability vectors. Consequently the objective can be written as

$$2\rho C + \max_{p,q \in \Delta_N} \left\{ p^\top y^+ - q^\top y^- + [F^\top (q - p)]^\top \theta + \sum_k \lambda_k |\theta_k| \right\}.$$

For a scalar z,

$$\min_{|v|\le B}\{zv+\lambda|v|\} = -B(|z|-\lambda)_+.$$

Indeed zero is minimizing when $|z| \le \lambda$, whereas the appropriate endpoint of the coefficient interval is minimizing otherwise. Applying this identity coordinatewise proves weak duality in (23). The finite program is feasible and has a finite attained optimum. Linear-programming duality, or equivalently minimax on the compact coefficient box and the two probability simplexes, interchanges the minimum and maximum in the last display. It gives equality for some p, q. For any explicitly supplied feasible triple, subtracting its dual lower bound from its primal objective bounds its excess over the optimum. *Q.E.D.*

Rational verification and the backward qualification chain. The empirical implementation uses rational feature values, rational label endpoints and rational moduli. A numerical optimizer proposes coefficients and dual multipliers. Coefficients are rounded to a fixed dyadic lattice and clipped to the declared box. Nonnegative dual proposals are rounded and normalized to sum exactly to one; a fixed unit vector is the valid fallback if their mass vanishes. The objective, support function and optimization gap are then recomputed in rational arithmetic. Solver failure or a gap exceeding the declared tolerance is recorded rather than converted into an assertion of optimal fitting.

At date t this argument conditions on the actual h_{t+1} produced by the next date's computation. Repeating backwards gives valid bounds $\omega_t \le \mathcal{W}_t$ simultaneously because they are deterministic qualifications, not confidence statements. Substituting them into the frozen-continuation proof gives

$$\Omega_h \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T, \qquad J_0^\pi - V_0^* \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T + H_h^\pi.$$

The same substitution is valid in the cost-contrast support. This does not compare the future continuations generated by different backward optimization choices, and hence makes no joint optimality assertion.

6.2. *Whole-cell policy qualification in the executed construction*

For the new two-state construction, the original terminal payoff is retained exactly. Each subsequent continuation is additive with nine coefficients: a constant and four features per state coordinate. The ReLU family uses x_j and $(x_j - k/4)_+$, $k = 1,2,3$. The conventional family uses the four nonconstant degree-four Bernstein terms $\binom{4}{k} x_j^k (1-x_j)^{4-k}$, $k = 1,\dots,4$. Feature Lipschitz bounds are respectively one and four. The latter follows by differentiating a Bernstein term: its derivative is four times the difference of two nonnegative degree-three Bernstein terms whose sum over the full basis is one. Both families have the same coefficient box and node information; their spanning spaces and modulus bounds need not coincide.

The midpoint cover with n cells per coordinate has radius $\rho = 1/n$ in ℓ^1. If the actual future has Lipschitz constant L_+, the inherited primitive inequalities give

$$L_x = 4 + \beta \frac{11}{16} L_+, \qquad L_a = \frac{13}{16} + \beta \frac{3}{4} L_+, \qquad C = L_x + \frac{L_a}{16}, \quad \beta = \frac{15}{16}.$$

At the terminal continuation $L_+ = 5$; at an earlier date it is the verified coefficient-weighted feature modulus. The 33-point action mesh at a midpoint x has exact fill radius $c(x)/64$. Thus subtracting $L_a c(x)/64$ from its minimum action value gives the lower continuous-action label, while the minimum itself is the upper label. All those evaluations are rational conditional expectations under the original uniform innovation law.

For the implemented actor, a second 33-point menu is rounded down to the $1/4096$ lattice and bounded by the minimum capacity over the whole observation cell. Let a_j minimize its exact midpoint action value and let y_j^- be the continuous-action lower label. At any state x in that cell,

$$\begin{aligned} Q^h(x,a_j) - \inf_a Q^h(x,a) &\le Q^h(x_j,a_j) - y_j^- + (L_x + C)\|x - x_j\|_1 \\ &\le Q^h(x_j,a_j) - y_j^- + (L_x + C)\rho. \end{aligned}$$

Taking the maximum over all cells gives the recorded uniform greedy-gap allowance. Replacing a_j by zero gives the comparator allowance. The selected action is feasible throughout its cell, including its closed boundary, and a fixed boundary

convention gives a well-defined actor. This proves the continuous-state, continuous-action qualification; midpoint evaluations alone would not do so.

For a ReLU future, exact conditional integration follows Proposition 7. The independent checker instead splits the innovation interval at every hinge crossing and integrates each affine piece by its endpoint trapezoid. For a Bernstein future, the production calculation expands the uniform moments. The independent checker uses the positive five-point weights $(7,32,12,32,7)/90$ on equally spaced points from $-1/32$ to $1/32$. Substitution of each monomial of degree zero through five verifies this rule directly, so it integrates the conditional degree-four polynomial exactly. The checker separately uses the retained original rational terminal primitive at every final-date action value. These checks concern all new construction nodes and actions, not the entire historical recursive Bellman computation.

6.3. *A time-uniform endpoint bound*

PROOF OF THEOREM 29: Condition on $\mathcal{F}_{i-1}$. The frozen-continuation telescoping proof applies to the two complete controllers selected for this episode; it gives $\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] = \Delta_i$. History-dependent actions cause no change in the tower property provided the actual conditional transition kernels and all required greedy-gap qualifications remain valid. The common initial law cancels the initial continuation term. The terminal correction is still required.

Write $R_i = b_i - a_i$. Since $L_i \le D_i \le U_i$ and the endpoints lie in $[a_i, b_i]$,

$$\mathbb{E}[L_i \mid \mathcal{F}_{i-1}] \le \Delta_i \le \mathbb{E}[U_i \mid \mathcal{F}_{i-1}].$$

Conditional Hoeffding's lemma therefore gives, for each fixed $\lambda > 0$,

$$\mathbb{E}\left[\exp\left\{\lambda(L_i - \Delta_i) - \frac{\lambda^2 R_i^2}{8}\right\}\middle|\mathcal{F}_{i-1}\right] \le 1.$$

To see the constant directly, a variable supported on an interval of length R has tilted variance at most $R^2/4$. The second derivative of its centered log moment-generating function is this tilted variance. Integrating twice from zero bounds that log moment-generating function by $\lambda^2 R^2/8$. The nonpositive conditional mean of

$L_i - \Delta_i$ can only decrease the exponential bound. The same reasoning applies to $\Delta_i - U_i$.

Products of these conditional factors are nonnegative supermartingales starting at one. Ville's inequality bounds the probability that the lower-endpoint product for λ_j ever exceeds $2/(\alpha w_j)$ by $\alpha w_j/2$. The same bound holds for the upper-endpoint product. A union bound over both signs and all j gives simultaneous validity with failure probability at most $\alpha \sum_j w_j \le \alpha$. Outside that exceptional event every partial sum is bounded by every corresponding linear boundary. Taking their minimum gives (77) and (78). The predictable support also bounds each Δ_i, justifying the additional intersection.

The event already covers every finite index, so evaluating it at a finite stopping time does not consume additional error probability. With a fixed complete controller pair and fixed initial-law/reset contract, its conditional expected original-law contrast is the same number Δ at every episode. Predictable changes of a valid critic alter the observations and ranges, not this identity. Without that fixed target, the preceding calculation identifies only the displayed conditional average. *Q.E.D.*

The implementation need not evaluate exponentials. For $x \ge 1$, write $x = 2^k y$ with $1 \le y < 2$, and use

$$\log y = 2 \sum_{r=0}^{m-1} \frac{z^{2r+1}}{2r+1} + \mathcal{R}_m, \qquad 0 \le \mathcal{R}_m \le \frac{2z^{2m+1}}{(2m+1)(1-z^2)}, \quad z = \frac{y-1}{y+1}.$$

Use the same expression at $z = 1/3$ for $\log 2$. It gives an exact rational upper bound on each logarithm in the boundary; replacing the logarithm by that bound enlarges the interval. Arithmetic operation counts do not bound rational bit complexity.

The finite set of bets is deliberate: a finite minimum is directly computable. It remains time-uniform for arbitrarily many episodes, although a finite bet set can leave a positive asymptotic radius. Increasing the predeclared set improves its supported scales but is a new design choice, not a free after-the-fact selection. One may instead preallocate a countable summable family; an implementation using any finite subset retains validity with its assigned weights.

6.4. *Stopping, failed episodes and the replacement decision*

PROOF OF COROLLARY 30: On the simultaneous event, $l_N \le \Delta \le u_N$. Adding the declared incremental charge and the one-time charge divided by the deployment count gives the claimed interval. Its upper endpoint being negative implies strict net cost reduction. Inclusion in an equivalence band is a different two-sided assertion. The event supports every such deterministic algebraic implication, including a stopping rule that inspects the interval repeatedly.

For a new epoch, condition on the history at its start, which includes the construction and selection of its new controller pair. Apply the theorem with that epoch's previsible error allocation α_e. A conditional union bound followed by expectation bounds the probability of any erroneous qualification by $\mathbb{E}\sum_e \alpha_e$, no larger than the total declared budget. No independence between epochs is required. *Q.E.D.*

A support reserved before an episode remains valid when its numerical evaluation fails. Recording the full interval $[a_i, b_i]$ then preserves endpoint containment and the martingale argument. Deleting the episode after an outcome-dependent failure need not preserve the conditional law. All attempted work is retained irrespective of which interval is recorded. An invalid support certificate is different: it supplies no admissible substitute observation and must not be treated as a harmless numerical timeout.

Finally, a conditional kernel-bias bound can be accommodated only when it is itself justified: if $|\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] - \Delta_i| \le \zeta_i$ with previsible certified ζ_i, enlarge both ends by $n^{-1}\sum_{i\le n}\zeta_i$. This follows by applying the proved bound to the actual conditional mean and then using the triangle inequality. An empirical model discrepancy without such a bound is not set to zero.

7. PROOFS FOR CONSTRUCTED-POLICY COST QUALIFICATION

PROOF OF (81) AND THE OBSERVATION IDENTITIES: Write the final state as $(m_1 + w, m_2 - w)$, where $\mathbb{E}w = 0$ and $\mathbb{E}w^2 = 1/3072$. The sum of the coordinates, and hence the shortage term, is constant in w. The two state-square terms in g contribute $4\mathbb{E}w^2$; the coordinate-difference square contributes $\mathbb{E}w^2$. All linear terms vanish in expectation. This proves (81). Since the first two actions are measurable

with respect to the initial state and the first innovation, conditional expectation of (79) with respect to those variables is (80). The tower property gives the equal expected costs. Applying the conditional-variance decomposition to the paired cash difference gives the stated variance inequality, including when the two actors have different acquisition resolutions.

For the primitive support, the lower next-state bound is $1/32$. Termwise upper bounds give $29/32$ in the first coordinate and $27/32$ in the second, so the unit cube is invariant for every feasible action. On that cube, $|x_j - 5/8| \le 5/8$, $|x_1 - x_2| \le 1$, the positive shortage is at most $1/2$, and $a \le 1/4$. Substitution into the nonnegative cost terms gives $c \le 103/64$ and $g \le 37/16$. Discounting gives (82); conditional expectation preserves the same support. *Q.E.D.*

PROOF OF THEOREM 31: Condition on all information used to construct the fixed policies. Lower and upper endpoint variables are bounded functions of independent bin rows. Their expectations bracket the original estimand, by pointwise enclosure or by integration within each bin. Apply the one-sided bounded empirical-Bernstein inequality to U_i and to $-L_i$, after affine rescaling to an interval of length one. Each failure probability is at most $2e^{-\ell}$, with the radius in the theorem. A union bound over two endpoint events for each of K estimands gives $4Ke^{-\ell}$. It requires no independence among estimands on a shared row. Subtracting the lower confidence endpoint from the upper one yields (84). Enlarging either radius and rounding confidence endpoints outward cannot remove the target from a covering interval. Integrating over the conditioning information proves the unconditional assertion when the construction itself was random. *Q.E.D.*

Validity of the two numerical realizations. For binary64 operations, each computed elementary lower endpoint is moved to the adjacent lower representable number and each upper endpoint to the adjacent upper number. For integer-directed operations, multiplication on the 2^{-48} grid uses the floor of the smallest endpoint product divided by 2^{48} and the ceiling of the largest. Division by a positive integer uses the corresponding integer floor and ceiling. Addition and negation are exact on this grid. Squaring includes zero when its argument interval straddles zero, and the

positive-part map is monotone. These rules enclose the result of each elementary operation.

Induction through either polynomial formula therefore encloses the cost and transition of every point in the input box. At an acquisition step, each possible true state belongs to one of the enumerated intersected cells, so its deployed action lies between the enumerated extrema. A second induction through the two dates proves the full-bin cost enclosure. Conditional final integration is valid by the exact identity already proved. The state and cost intersections use established primitive supports. Taking the hull of two valid intervals is conservative even when one is wider. Neither interval intersection with a sampled value nor agreement at a midpoint is used to establish this full-bin claim.

In the statistical implementation, sample sums and sums of squares are integers. Their rational combination gives the unbiased variance exactly. For a rational $v \geq 0$, the least k satisfying $k^2 \geq 2^{120}v$ gives the upward square root $k/2^{60}$. Each radius enlargement is less than 2^{-60}, except that it is zero for an exactly represented root. The recorded width decomposition uses these enlarged roots and is an exact rational identity. The independent midpoint trajectories and synthetic boundary cases are regression checks supplementary to, not substitutes for, the enclosure argument.

8. PROOFS FOR COMPLETE-POLICY AND POPULATION TRANSFER

8.1. *Original-law occupation identity*

Fix $\theta \in \Theta_\epsilon$ and use the law generated by its complete feasible policy. Boundedness permits finite summation and conditional expectation. From $B_{t+1} = B_t\beta_t$ and (43),

$$\begin{aligned}\sum_{t<T} B_t \mathbb{E}_{\pi_\theta} A_t^\tau(X_t, \pi_{\theta,t}(X_t)) &= \sum_{t<T} B_t \mathbb{E}_{\pi_\theta} c_t(X_t, \pi_{\theta,t}(X_t)) \\ &\quad + B_T \mathbb{E}_{\pi_\theta} V_T^\tau(X_T) - \mathbb{E} V_0^\tau(X_0) \\ &= J_0^{\pi_\theta} - J_0^\tau.\end{aligned}$$

The conditional-expectation step uses the candidate's transition, not the reference occupation. The terminal equality uses the common original terminal cost $V_T^\tau = g$.

Define y_{tCa} as the indicator that this candidate selects label a in cell C, $m_{tC} = \Pr_{\pi_\theta}(X_t \in C)$, and $q_{tCa} = m_{tC} y_{tCa}$. Define z_{tCaD} as the joint probability of $X_t \in C$, label a, and $X_{t+1} \in D$. Then the initial, conservation and assignment equations in (46) follow directly. Integrating the pointwise transition bounds against the candidate's subprobability law on C gives the two inequalities for z, including when $q = 0$. An actually realized common parameter cannot realize all the events in a certified forbidden set, so every conflict inequality holds. Consequently this integral point belongs to $\mathcal{P}_\epsilon$. Each stage's advantage expectation is bounded by $\sum_{C,a} \ell_{tCa} q_{tCa}$ and $\sum_{C,a} u_{tCa} q_{tCa}$. Taking extrema over the outer polytope proves Theorem 14.

The polytope is closed and bounded: m_t, q_t are subprobabilities or probabilities, $0 \le y \le 1$, and z is bounded by its nonnegative row sum. It is nonempty whenever the qualification class is nonempty, in particular when it contains $\widehat{\theta}$. Hence its extrema exist. Adding valid constraints shrinks it. If a smaller tolerance retains all larger-tolerance constraints and possibly more, its lower extremum increases and its upper extremum decreases. Independent dual feasibility supplies the appropriate weak-duality bound even when the numerical solve is incomplete. Adding and subtracting the two reference costs proves the class comparison. These operations are deterministic on the simultaneous input event.

Why reference weights are insufficient. Consider a two-date, zero-stage-cost example. Initially only state s is visited. At date zero, action L moves to state ℓ and R to state r. At date one the costly action costs one at r and zero at ℓ. The reference selects L; a candidate selects R and then the costly action at r. The reference's date-one occupation gives r mass zero, but the candidate's cost difference is one. A first-date formula using the reference continuation sees zero date-zero contrast and omits the later change. The flow constraint instead forces $m_{1r} = 1$ and charges the date-one advantage. This example identifies an error that cannot be repaired by a union bound over cells.

Shared coefficients remain material. In the same example, let $\theta \in \{-1, 1\}$. At s action R has score $-\theta$ and L score zero; at r the costly action has score θ and the free action score zero. Each event is individually possible, but R and the costly action require opposite signs. Every common-parameter policy has cost zero, whereas

the unconstrained action relaxation permits cost one. The exact conflict inequality excludes their simultaneous integral choice. Its fractional relaxation can still admit cost $1/2$; this is why the theorem asserts validity and monotone tightening, not exact attainability of every relaxed occupation.

8.2. *Nonlinear evaluations and strategic deviations*

Assume inductively that $V^{\tau}_{t+1} + e^{-}_{t+1} \le V^{\pi}_{t+1} \le V^{\tau}_{t+1} + e^{+}_{t+1}$. Monotonicity and cash invariance give

$$\mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{-}_{t+1} \le V^{\pi}_{t} \le \mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{+}_{t+1}.$$

Subtract V^{τ}_{t} and add the stated residual bounds. The common terminal condition starts the induction. Under a general continuation Lipschitz modulus instead, triangle inequality gives $\|V^{\pi}_{t} - V^{\tau}_{t}\|_{\infty} \le b_t + \lambda_t \|V^{\pi}_{t+1} - V^{\tau}_{t+1}\|_{\infty}$; iteration proves the second formula of Proposition 15 on the stipulated invariant domains.

For player i, fix every other player's complete strategy. The induced controlled kernel and the player's feasible actions define a best-response problem, with the player's own stated recursive objective. Its best-response value is below the cost of the attained action sequence. Apply the one-sided residual sandwich to this problem from every date and state. A bound r_{it} on its difference is exactly a bound on that player's profitable unilateral cost reduction. Applying it simultaneously to every player and subgame establishes the claimed approximate deviation property. This argument does not establish existence of a profile satisfying the premises. For temporal selves, each self is a separate player with its own deviation problem; no identity with a commitment planner's optimum is asserted.

If a numerical kernel $\widetilde{\mathcal{T}}^{\pi}_{t}$ differs uniformly from the original kernel by at most κ_t on the continuations in use, it adds κ_t to the stage residual. Its contribution is therefore at most $\sum_t (\prod_{j<t} \lambda_j)\kappa_t$. To compare a numerical policy with the original optimum, the corresponding optimal-operator approximation must also be charged in the two-sided sandwich. This explicitly accounts for a diffusion approximation without postulating an unproved rate in its time step.

9. PROOFS FOR JOINT-DOMAIN CONSTRUCTION AND EXPLORATION

9.1. *One population circuit*

Let $G_t(r) = \mathcal{T}_t f_{t+1}(r)$ on the joint domain. Primitive regularity and the Hausdorff action bound imply that it is L_t-Lipschitz by the same comparison as in Theorem 2, now including the context coordinates. For each node, $-e_t \le y_{ti} - G_t(r_{ti}) \le D_t k_t + e_t$. Thus every cone is at least $G_t(r) - e_t$, and a node within h_t gives a cone at most $G_t(r) + D_t k_t + e_t + 2L_t h_t$. The lower envelope has Lipschitz modulus L_t because each of its members does. The original label owner, the feasible repair, and the original action modulus give the selected-policy upper residual $e_t + D_t \zeta_t$ exactly as in the original construction. The argument closes backward from the terminal approximation, and then the original policy sandwich proves the loss account.

Use $|u| = \mathrm{ReLU}(u) + \mathrm{ReLU}(-u)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$. Compute all cone values and combine them in a balanced minimum tree. This proves the claimed weight/operation count and depth, up to the fixed affine layers. Lexicographic original-owner selection is a separate measurable comparison operation; the discontinuous actor is not falsely called a continuous ReLU function. Its feasible repair is charged separately. The guarantee holds at every point of every joint domain, so evaluating it along an endogenous sequence uses no distributional inference or independence assumption. The covering exponent and effective construction cost remain explicit.

9.2. *Adaptive acquisition coverage*

Fix one cover ball B. Let E_i be the event that the first i acquisitions miss it. By the conditional lower-probability assumption,

$$\Pr(E_i) = \mathbb{E}[\mathbf{1}_{E_{i-1}} \Pr(Y_i \notin B \mid \mathcal{F}_{i-1})] \le (1 - a_h) \Pr(E_{i-1}).$$

Induction yields $(1 - a_h)^n$. A union bound over the M fixed balls gives the result. If none is missed, any point is within h of a cover center and that center is within h of an acquired node, hence the radius is $2h$. This argument controls geometry, not label error; any probabilistic label-certification event is included separately in the

overall error allocation. Pure exploitation without a conditional lower-probability bound does not satisfy the proposition.

9.3. *Prediction and fallback work*

Suppose the best action is a_* and $\Delta > 4e$. For every competitor b, its estimate exceeds that of a_* by at least $\Delta - 2e > 2e$. The estimate-minimizing action is consequently a_*, and its upper confidence endpoint is below every competitor's lower endpoint. Thus failure of the screen implies $\Delta \le 4e$. Its probability is at most the tie mass plus $C(4e)^\alpha$, truncated at one. Multiplying by the fallback's upper work bound and adding proposal and test overhead proves Proposition 19. To certify a saving one must compare that upper account with a lower account for the conventional baseline, not with another upper bound. This distinction is the work analogue of not subtracting policy-loss upper bounds to estimate a cost difference.

10. PROOF OF THE DATED ACCURACY ALLOCATION

Hölder's inequality gives, for every feasible positive allocation,

$$\begin{aligned} S &= \sum_t (a_t \eta_t^{-p})^{1/(p+1)} (w_t \eta_t)^{p/(p+1)} \\ &\le \left(\sum_t a_t \eta_t^{-p} \right)^{1/(p+1)} \left(\sum_t w_t \eta_t \right)^{p/(p+1)}. \end{aligned}$$

Therefore the objective is at least $S^{p+1} \varepsilon^{-p}$. Substituting (72) attains equality and uses the whole error budget. Strict convexity of $\eta \mapsto \eta^{-p}$ gives uniqueness. The discounted expression follows by summing the geometric series with ratio $\beta^{p/(p+1)}$; the undiscounted expression follows from $S \le (T+1) A^{1/(p+1)}$. For the clipped allocation $\eta'_t = \min(\eta_t^*, \eta_{\max})$, the error budget can only decrease and

$$a_t (\eta'_t)^{-p} = a_t \max\{(\eta_t^*)^{-p}, \eta_{\max}^{-p}\} \le a_t (\eta_t^*)^{-p} + a_t \eta_{\max}^{-p}.$$

The asserted additive bound follows. Dyadic downward rounding by less than a factor two proves the final factor. Bit complexity, per-date setup and storage costs lie outside this algebraic upper model unless expressly included in a_t.

11. PROSPECTIVE MENU CHOICE AND UNIFORM ECONOMIC SENSITIVITY

11.1. *Finite-menu selection*

Condition on everything used to construct the finite service menu. Let $C_j(W) \in [0, H]$ be the true cost of service j on a new IID workload W, and suppose $l_{ji} \le C_j(W_i) \le u_{ji}$ for each evaluated workload. The endpoints are valid whole-input enclosures, not unqualified point calculations. A bounded-variable exponential inequality and a union bound over the m services give, with probability at least $1 - \delta$,

$$|\overline{C}_j - J_j| \le q_n, \quad q_n = H\sqrt{\log(2m/\delta)/(2n)}, \quad j = 1, \dots, m.$$

For completeness, the conditional moment-generating-function bound for a centered variable in an interval of length H is at most $\exp(\lambda^2 H^2/8)$. Iterating it over the independent draws, applying Markov's inequality and optimizing λ proves each tail $\exp(-2nq_n^2/H^2)$. Hence $J_j \in [\bar{l}_j - q_n, \bar{u}_j + q_n]$ simultaneously.

Let service $\widehat{j}$ minimize $\lambda\bar{u}_j + p \cdot w_j + f_j$ for $\lambda \ge 0$, and let j_* minimize its true counterpart. On this event, the selected objective minus the true optimum is at most

$$\begin{aligned} &\lambda(\bar{u}_{\widehat{j}} + q_n) + p \cdot w_{\widehat{j}} + f_{\widehat{j}} - \lambda(\bar{l}_{j_*} - q_n) - p \cdot w_{j_*} - f_{j_*} \\ &\quad \le \lambda\{2q_n + \bar{u}_{j_*} - \bar{l}_{j_*}\} \le \lambda\{2q_n + \max_j(\bar{u}_j - \bar{l}_j)\}. \end{aligned}$$

The same event works for every price and fee vector simultaneously, because it was constructed for the costs before prices were applied. This does not make an outcome-dependent acquisition or construction of new services free of a selection charge. On a separate evaluation stream, simultaneous pairwise intervals give the sharper evaluated regret bound $\max\{0, \max_j u_{\widehat{j},j}\}$ for the calibration-selected service. The stream need not show strict superiority to establish noninferiority at a prespecified margin.

For adaptively generated workloads, iteration of the conditional moment-generating-function bound instead bounds the average predictable cost $n^{-1}\sum_i \mathbb{E}[C_j(W_i) \mid \mathcal{F}_{i-1}]$, for a menu fixed before that sequence. It does not identify a stationary population expectation or the cost under another policy's counterfactual

workload distribution. This is the correct distinction between sequential validation and an IID new-task claim.

11.2. *Price and fee regions*

For a compact joint cost polytope $\mathcal{D}$ and two feasible assignments x, y, the largest objective difference is

$$\sup_{c\in\mathcal{D}}\{\lambda c\cdot(x-y)+p\cdot(w_x-w_y)+f\cdot(x-y)\}$$
$$=\lambda h_{\mathcal{D}}(x-y)+p\cdot(w_x-w_y)+f\cdot(x-y).$$

Nonnegativity of λ is used here. The candidate x is optimal for every $c\in\mathcal{D}$ precisely when this expression is nonpositive for every feasible y. This proves the polyhedral region in Proposition 34; a certified support upper bound gives a sufficient subregion. Linear optimization over a compact price-fee polytope reduces uniform verification to its vertices. The boundaries are the stated price-fee breakpoints. If $\mathcal{D}$ includes correlated pairwise constraints they must be retained in its support problem; independently choosing the worst endpoint of each pair can be looser. No step estimates a price, monetary scale or adoption fee from a clock measurement.

12. PROOF OF PATTERN-CONSISTENT TRANSFER

12.1. *Embedding and comparison with the earlier relaxation*

Fix an actual qualified parameter θ and its complete pattern σ. Give that pattern weight one and all others weight zero. Under the original law of its complete policy, set $m^{\sigma}_{tC}=\Pr(X_t\in C)$ and $z^{\sigma}_{tCD}=\Pr(X_t\in C, X_{t+1}\in D)$. Integrating the whole-cell transition bounds gives (48). The same occupation identity proved in Supplement 8 bounds the true cost difference between the two objectives in (49). This proves validity for the entire parameter class. There is no additional probability union over its members.

To compare the two relaxations, take any disaggregated feasible point and define

$$y_{tCa}=\sum_{\sigma:\sigma(t,C)=a}\lambda_{\sigma},\qquad q_{tCa}=\sum_{\sigma:\sigma(t,C)=a}m^{\sigma}_{tC},$$

$$z_{tCaD} = \sum_{\sigma:\sigma(t,C)=a} z^{\sigma}_{tCD}, \qquad m_{tC} = \sum_{\sigma} m^{\sigma}_{tC}.$$

Conservation implies $\sum_C m^{\sigma}_{tC} = \lambda_\sigma$ at every date, hence $0 \le m^{\sigma}_{tC} \le \lambda_\sigma$ and $q_{tCa} \le y_{tCa}$. All assignment, flow and transition constraints of (46) follow by summation. Each retained pattern satisfies every certified conflict, so their convex combination satisfies its corresponding inequality in y. Both advantage objectives are unchanged by this projection. Consequently the disaggregated minimum is no smaller, and its maximum is no larger, than the earlier pair. Deleting only patterns certified impossible preserves coverage and shrinks the feasible set. Nonempty compactness follows from the actual-policy embedding, bounded total mass and closed linear constraints. This also proves existence of the displayed extrema and justifies ordinary linear-programming dual certificates.

12.2. *Quantitative comparison with an attained pattern*

Now assume every retained pattern is actual and has a qualification witness. Consider one pattern with positive weight and divide its flow variables by that weight. Its row sums define a time-inhomogeneous stochastic matrix $\widetilde{P}_t$ on the dated cells. At a row with zero mass choose any stochastic row within the same transition intervals; such a row exists because the original transition at any state in the nonempty cell supplies one.

Let m_t be the true cell law of this pattern's original continuous-state policy, and P_t its conditional cell-to-cell probabilities given the date-t cell under that law. The law inside a cell can depend on the complete earlier history. Nevertheless, conditional expectation and the whole-cell hypothesis put every row of P_t inside the same interval box. At an unvisited cell choose an arbitrary original-law row. This choice does not affect $m_t P_t = m_{t+1}$. No Markov assumption on the observed cell process has been imposed.

Two stochastic rows in this box differ in ℓ^1 norm by at most δ_t. Writing $\widetilde{m}_t$ for the normalized relaxed law, stochastic contraction gives

$$\|\widetilde{m}_{t+1} - m_{t+1}\|_1 \le \|(\widetilde{m}_t - m_t)\widetilde{P}_t\|_1 + \|m_t(\widetilde{P}_t - P_t)\|_1$$

$$\leq \|\widetilde{m}_t - m_t\|_1 + \delta_t.$$

The initial laws agree. Since the distance between two probability vectors is at most two, induction bounds this distance by e_t.

Let a_{tC} denote the true conditional expected reference advantage under this pattern at date t, with an arbitrary value in the enclosure on unvisited cells. Then $\ell_{tC\sigma(t,C)} \leq a_{tC} \leq u_{tC\sigma(t,C)}$. For the upper objective at that date,

$$\begin{aligned}\sum_C u_{tC\sigma(t,C)}\widetilde{m}_{tC} - \sum_C a_{tC}m_{tC} &= \sum_C u_{tC\sigma(t,C)}(\widetilde{m}_{tC} - m_{tC}) \\ &\quad + \sum_C (u_{tC\sigma(t,C)} - a_{tC})m_{tC} \\ &\leq M_t e_t + \omega_t.\end{aligned}$$

The lower objective is treated with the reverse inequality. Summing with B_t bounds any normalized pattern-flow objective within $\mathcal{E}$ of that pattern's true cost difference. Weighting by λ_σ and summing proves that every feasible upper objective is at most $j_+ + \mathcal{E}$ and every feasible lower objective at least $j_- - \mathcal{E}$. The actual-policy embeddings give the other inequalities in (50).

When the transitions and advantages are exact on a finite state space, $\delta_t = \omega_t = 0$. Each fixed pattern has a unique occupation sequence; the convex mixture of their costs has the same minimum and maximum as this finite set. A retained qualification witness therefore attains each extremal policy value. More generally the quantitative inequalities prove the stated convergence whenever $\mathcal{E}$ tends to zero uniformly. This is not a claim that arbitrary partitions, discontinuous deterministic transitions or incomplete pattern searches satisfy that condition.

12.3. *Strict improvement in the shared-sign example*

Let $\theta = 1$ select the right-hand initial action and the free action at the right-hand successor. Let $\theta = -1$ select the left-hand initial action and the costly action at the right-hand successor. All other costs are zero. Both complete policies cost zero. In the conflict relaxation, set the right-hand initial assignment and occupation to $1/2$, and the right-hand costly assignment and occupation also to $1/2$. Put the remaining

initial mass on the left-hand action; choose the free action in the left-hand successor. The forbidden pair's assignments sum to one, so its conflict inequality holds. Its cost is $1/2$. Conversely that cost cannot exceed either member of the forbidden pair, whose assignments sum to one, and thus is at most $1/2$. Pattern disaggregation has only the two zero-cost occupation sequences and gives upper bound zero. The numerical test computes the three bounds by independent path enumeration and linear programming.

13. PROOF OF COUPLED ACCURACY ALLOCATION

Write $f_j(\eta) = c_j \prod_t \eta_t^{-a_{jt}}$. For any vector v and positive η, direct differentiation yields

$$v^\top \nabla^2 f_j(\eta) v = f_j(\eta) \left\{ \left(\sum_t a_{jt} \frac{v_t}{\eta_t} \right)^2 + \sum_t a_{jt} \left(\frac{v_t}{\eta_t} \right)^2 \right\} \geq 0.$$

Because every coordinate appears with positive exponent somewhere, summing these forms is strictly positive for every nonzero v. Hence F is strictly convex on the positive orthant. Its feasible set is convex and nonempty. Every coordinate is bounded above by its cap and by ε / w_t. If a feasible sequence sends a coordinate to zero, a term containing that coordinate tends to infinity: all other coordinates in that term have fixed finite upper bounds. A finite sublevel set containing any feasible allocation is consequently bounded away from zero and compact. Continuity gives attainment; strict convexity gives uniqueness.

After $\eta_t = \exp(x_t)$, every nonconstant term is an exponential of an affine function. The budget becomes $\sum_t w_t \exp(x_t) \leq \varepsilon$ and each cap becomes $x_t \leq \log \bar{\eta}_t$. Both the objective and constraints are convex. This proves the alternative representation, without asserting that a floating-point optimization status is a proof.

At any feasible candidate, the supporting-hyperplane inequality is

$$F(\eta) \geq F(\eta^0) + g \cdot (\eta - \eta^0).$$

If $r_t = g_t + \lambda w_t + \nu_t \geq 0$, positivity of η and its budget and caps imply

$$g \cdot \eta = \sum_t r_t \eta_t - \lambda \sum_t w_t \eta_t - \sum_t \nu_t \eta_t$$

$$\geq -\lambda\varepsilon - \sum_t \nu_t \bar{\eta}_t.$$

Substitution proves (76). A sufficiently small positive vector satisfies all inequalities strictly, so Slater's condition holds. At the optimum the usual stationarity and complementary-slackness equations provide nonnegative λ, ν with $r = 0$, $\lambda(\sum_t w_t \eta_t - \varepsilon) = 0$ and $\nu_t(\eta_t - \bar{\eta}_t) = 0$. They make the lower certificate equal to the feasible objective. Equivalently, these conditions themselves give a complete sufficiency proof by the supporting-hyperplane inequality. Since $g_t < 0$, the stated maximum-ratio choice with $\nu = 0$ is always a nonnegative feasible certificate.

For integer exponents the objective, derivatives, constraints and certificate at rational arguments use only rational operations. The proof does not license substituting unqualified decimal derivatives when real exponents are used; outward enclosures must then make the claimed inequalities valid. For example, a rational affine lower support with verified slope bounds on the feasible box can replace the exact tangent. The finite fixtures use integer exponents and independently recompute the gradients, so no such analytic-rounding issue is hidden in their reported gaps.

If $\eta_t^0/2 < \eta_t^{\mathrm{dyad}} \leq \eta_t^0$, then

$$f_j(\eta^{\mathrm{dyad}}) \leq 2^{\sum_t a_{jt}} f_j(\eta^0) \leq 2^P f_j(\eta^0).$$

Summation proves the rounding bound, while decreasing coordinates preserves both the error budget and all upper caps. Finally, the equal allocation $\eta_t = \varepsilon/T$ is feasible under the corollary's hypotheses. Since $T/\varepsilon \geq 1$, every monomial is at most $C(T/\varepsilon)^P$. There are at most KT of them. Adding F_0 proves Corollary 28. Neither this feasible upper construction nor the dual lower bound on its resource model is an information-theoretic lower bound for all Bellman algorithms.

14. PROOFS FOR ORIGINAL-LAW TRANSPORT QUALIFICATION

14.1. *One-step comparison*

For probability measures μ, ν and a K-Lipschitz function f, every coupling Γ of the measures satisfies

$$\left| \int f \, d\mu - \int f \, d\nu \right| = \left| \int (f(y) - f(z)) \, \Gamma(dy, dz) \right| \leq K \int d(y,z) \, \Gamma(dy, dz).$$

Taking an infimum over couplings proves the required KW_1 bound. The argument does not require densities or attainment of the infimum. Boundedness or the assumed finite first moments makes the integrals well defined.

Fix a qualified θ and suppress its subscript. Define the original-law action operator

$$Q_t^P f_{t+1}(x,a) = c_t(x,a) + \beta_t P_t^a f_{t+1}(x).$$

The preceding coupling inequality and (51) give, for every original feasible action,

$$|Q_t^P f_{t+1}(x,a) - \widehat{Q}_t(x,a)| \leq b_t. \tag{4}$$

If two functions on a nonempty action set differ uniformly by at most b, their infima differ by at most b: each function is everywhere at least the infimum of the other minus b, and the reverse argument gives the upper bound. No differentiability, uniqueness or continuity of the minimizing action is needed. It follows that

$$\begin{aligned} \ell_t - b_t &\leq T_t f_{t+1} - f_t \leq u_t + b_t, \\ \ell_t - b_t &\leq T_t^\pi f_{t+1} - f_t \leq u_t + \gamma_t + b_t. \end{aligned} \tag{5}$$

The lower policy inequality uses feasibility: the approximate action value at the implemented action is at least its infimum. The upper inequality uses the certified approximate greediness. All inequalities hold on the whole original state domain.

14.2. *Backward induction and the original optimum*

For a sequence r_t, define $S_t(r,d) = \sum_{s=t}^{T-1} (\prod_{j=t}^{s-1} \beta_j) r_s + (\prod_{j=t}^{T-1} \beta_j) d$. Empty products equal one. The claim

$$S_t(\ell - b, d^-) \leq V_t^* - f_t \leq S_t(u + b, d^+),$$

$$S_t(\ell - b, d^-) \le V_t^\pi - f_t \le S_t(u + \gamma + b, d^+)$$

follows by backward induction. At T, it is the terminal enclosure. Suppose the first inequality holds at $t+1$. Order preservation and the identity $T_t(v+k) = T_t v + \beta_t k$ give its lower bound at t from the lower bound in (5). The upper bound follows in the same way. The fixed-policy operator has the same order and constant-shift properties, so the second line follows separately. This induction is under the original kernels; the numerical kernels do not determine the economic value being bounded.

Subtract the lower bound for $V_0^* - f_0$ from the upper bound for $V_0^\pi - f_0$. The result is exactly $\mathcal{R}_\theta$. Feasibility gives $V_0^\pi \ge V_0^*$ and proves (53). The proof holds at each initial state and hence after averaging under any initial distribution supported on the declared domain. It also holds for every parameter for which the uniform hypotheses were certified. If the hypotheses are established on a single simultaneous data event, the conclusion holds on that event without an additional union over parameter vectors. Pointwise certificates obtained only at observed states do not create that simultaneous event.

The parameter may determine a different future continuation and a different future action at every date. We fixed it only to prove an inequality for its complete policy; the proof never substitutes another parameter's future. Discontinuous acquisition or deterministic tie-breaking therefore causes no problem as long as the resulting policy is measurable, feasible and satisfies the whole-domain residual inequalities. A transition bound applying only under the reference policy would not establish (4) and is insufficient.

14.3. *State-independent shifts and singular laws*

Replace f_t by $f_t + k_t$ and $\widehat{Q}_t$ by $\widehat{Q}_t + \beta_t k_{t+1}$. Since both kernels have total mass one, the arithmetic discrepancy, transport modulus and greedy gap are unchanged. The approximate Bellman residual is shifted by $\beta_t k_{t+1} - k_t$. Its certified lower and upper endpoints shift by that same amount, so their difference is unchanged. The terminal enclosure endpoints both decrease by k_T. Every term of $\mathcal{R}_\theta$ is consequently invariant. Using absolute levels of the continuation in place of its oscillation would forfeit this property.

If $\widehat{P}_t$ is the law of $q_t(Y)$ when Y has law P_t, the joint law of $(Y, q_t(Y))$ is a coupling. Its expected distance is at most h_t. Thus $W_1(P_t, \widehat{P}_t) \le h_t$. Substitution into $2\sum_t B_t b_t$ gives (54). In particular, δ_h and δ_0 have transport distance h but total-variation distance $\sup_A |\delta_h(A) - \delta_0(A)| = 1$ when $h > 0$. Mutual singularity therefore does not obstruct this continuation-based calculation. The conclusion concerns the transport contribution only; a nonvanishing residual span or greedy error still has to be paid.

14.4. *Stationary discounted bound*

Let T and T^π be the bounded discounted Bellman and policy operators. They preserve order, translate constants by β and are sup-norm contractions with modulus β. Iterating (5) from the bounded function f, geometric summation yields

$$\frac{\ell - b}{1-\beta} \le V^* - f \le \frac{u + b}{1-\beta}, \qquad V^\pi - f \le \frac{u + \gamma + b}{1-\beta}.$$

The contraction limits are the original optimal and policy values. Alternatively, apply the finite-horizon induction and let the bounded discounted terminal difference tend to zero. Subtraction proves Corollary 21. This argument requires a certified modulus of the stored f; it does not infer a uniform modulus for every optimal value from Lipschitz primitives. That distinction matters when transition dynamics expand distances.

14.5. *Resource allocation and the other economic objectives*

With $\eta_t \ge u_t - \ell_t + \gamma_t$, the sufficient budget (55) is obtained by rearranging the nonnegative terms in $\mathcal{R}_\theta$. After subtracting fixed terminal and arithmetic contributions, it is a linear budget in the residual and transport allowances. On positive-weight coordinates, any resource upper model satisfying Theorem 27 has that theorem's convexity, attainment, dyadic-rounding and dual-certificate properties. This is a direct application with additional accuracy coordinates, not a newly asserted lower bound on all numerical methods. Caps and certificates for the moduli are part of the input. Optimizing a modulus jointly requires retaining its parameter dependence in the resource and error constraints.

For a cash-invariant monotone one-step risk operator with discount β_t, a certified numerical discrepancy b_t gives the same two residual inequalities and the same constant-shift induction. Neither the coupling proof for linear expectation nor cash invariance alone supplies that numerical discrepancy for a nonlinear risk functional. For a general Lipschitz aggregator, the separate absolute-error induction in Proposition 15 supplies the valid replacement. Applying an original-optimum bound to each fixed-opponent best-response problem gives the stated unilateral incentive account; it does not prove existence of a game equilibrium. These arguments preserve the distinct hypotheses of the original applications.

14.6. *Diagnostic with a falsely free transition approximation*

In the one-date example, both numerical action values are constants: zero for action 0 and $h/2$ for action 1. Hence $\widehat{T}f_1 - f_0 = 0$, the numerical greedy gap is zero, and the terminal enclosure is exact. The true values of the two actions are h and $h/2$, respectively. The numerical greedy policy therefore has regret $h/2$. The stored terminal function has modulus one and the largest transport discrepancy is h, so Theorem 20 gives $2h$. The omitted-charge expression is zero and is invalid. All assertions are exact for rational h; no sampling or floating-point limit is used.

15. SIGNED QUALIFICATION AND NONLINEAR TRANSPORT: PROOFS

15.1. *Proof of Theorem 22*

Fix θ and omit it from the notation. Because b_t^- is valid for every original feasible action, taking infima gives

$$T_t f_{t+1} - f_t \geq \widehat{T}_t - f_t + b_t^- \geq \ell_t + b_t^-.$$

For the actual feasible action, the numerical greediness and actor-specific upper bounds give

$$T_t^{\pi} f_{t+1} - f_t \leq \widehat{Q}_t(\cdot, \pi_t) - f_t + b_t^{\pi,+} \leq u_t + \gamma_t + b_t^{\pi,+}.$$

Define real constants backwards by

$$a_T = d^-, \quad c_T = d^+, \qquad a_t = \ell_t + b_t^- + \beta_t a_{t+1}, \quad c_t = u_t + \gamma_t + b_t^{\pi,+} + \beta_t c_{t+1}.$$

At the terminal date $f_T + a_T \leq g \leq f_T + c_T$. If $V^*_{t+1} \geq f_{t+1} + a_{t+1}$, monotonicity and cash invariance, including under the infimum, imply

$$V^*_t = T_t V^*_{t+1} \geq T_t f_{t+1} + \beta_t a_{t+1} \geq f_t + a_t.$$

Similarly, $V^\pi_{t+1} \leq f_{t+1} + c_{t+1}$ implies

$$V^\pi_t = T^\pi_t V^\pi_{t+1} \leq T^\pi_t f_{t+1} + \beta_t c_{t+1} \leq f_t + c_t.$$

Induction proves both comparisons. Subtracting at date zero gives $V^\pi_0 - V^*_0 \leq c_0 - a_0$, which expands to (57). Feasibility and optimality give its lower bound zero. Infima need not be attained: the order comparison holds directly under infima, provided the stipulated original value recursion and feasible policy are defined. Every parameter is treated by the same argument; a supremum of valid bounds provides the stated classwise conclusion. No union over parameter vectors and no reference-policy occupation law is introduced.

For shifts $f'_t = f_t + k_t$, set $\widehat{Q}'_t = \widehat{Q}_t + \beta_t k_{t+1}$. Both numerical residual endpoints shift by $\beta_t k_{t+1} - k_t$; their difference is unchanged. Numerical action differences and original-minus-numerical action values are unchanged by cash invariance. Both terminal endpoints shift by $-k_T$. Thus every term in (57) is invariant. If the action-value discrepancy itself is a constant k_t, choose $b^-_t = b^{\pi,+}_t = k_t$ and its charge vanishes.

For the stationary assertion, the Bellman and fixed-policy maps are sup-norm contractions with modulus β: monotonicity and cash invariance imply sup-norm nonexpansiveness of each one-step evaluation. The finite-horizon argument with a bounded terminal comparison then has a terminal remainder of order β^T. Letting T tend to infinity proves the displayed stationary bound. The asserted constant-shift invariance does not justify shifting an original preference primitive or an action-dependent immediate cost.

15.2. *Sharpness and the distinction between constant and statewise bias*

For $h > 0$, take one decision date, terminal cost $g(y) = y$, discount one and zero immediate costs. The original successors of actions zero and one are respectively

h and zero; the numerical successors are both zero. Store $f_0 = 0$ and $f_1 = g$, and let the numerical tie convention select action zero. All numerical residual and greediness terms vanish. The original-minus-numerical action values lie in $[0, h]$, the selected-action upper endpoint is h, and the implemented policy loses exactly h. Thus the signed charge h is attained. The symmetric allowance $2h$ remains valid but is strictly larger. The example works for any monotone cash-invariant normalized risk evaluation because its successors are deterministic. Both laws can be singular, so no total-variation convergence is being used.

A statewise common-action bias is different. Let a root action choose state L or R at the next decision date. At that date there is only one feasible action, with original cost $2h$ in L and zero in R. Let every numerical cost, stored continuation and terminal cost be zero, with discount one. Break the root numerical tie toward L. All numerical residual spans and greediness gaps vanish. At the second date the discrepancy is common to every action at its state, but equals $2h$ or zero depending on that state. The implemented policy loses $2h$. Cancelling the discrepancy merely because it is action independent at each state would falsely report zero loss. The signed uniform range correctly charges $2h$. This is precisely the earlier action's effect on future states; fixing the reference occupation would also lose it.

15.3. *Proof of Proposition 23*

Let $(Y, \widehat{Y})$ have any coupling of the two successor laws. For every real threshold z, the map $r \mapsto (r - z)_+$ is one-Lipschitz. Hence

$$\left| P(f - z)_+ - \widehat{P}(f - z)_+ \right| \leq \mathbb{E}|f(Y) - f(\widehat{Y})| \leq K\mathbb{E}d(Y, \widehat{Y}).$$

The same bound, divided by $1 - \alpha$, is uniform in z for the two functions minimized in (58). Infima of uniformly close functions are close. Taking the infimum over couplings proves the first bound, with approximating couplings sufficient when an optimum is not attained. At $\alpha = 0$ the variational expression equals expectation: the expression is at least Pf and equality holds for any threshold no larger than the bounded loss's infimum.

To verify that the tail factor cannot be uniformly reduced, take $f(y) = y$ on $[0,1]$, $\widehat{P} = \delta_0$ and $P = (1-h)\delta_0 + h\delta_1$, with $0 < h \leq 1$ and $\alpha = 1 - h$. Then $K = 1$, $W_1(P, \widehat{P}) = h$, $\rho_P^\alpha(f) = 1$ and $\rho_{\widehat{P}}^\alpha(f) = 0$. The bound $Ke/(1-\alpha) = 1$ is attained. In particular, a bound uniform up to risk level one cannot be deduced from Wasserstein distance alone with a fixed coefficient.

For the entropic assertion, choose $m \leq f \leq M$. For positive λ, the mean value theorem gives

$$|\exp(\lambda f(y)) - \exp(\lambda f(\widehat{y}))| \leq \lambda \exp(\lambda M) K d(y, \widehat{y}).$$

Both exponential integrals are at least $\exp(\lambda m)$, and $|\log s - \log t| \leq |s - t| / \min(s,t)$ for positive s, t. Combining these facts, dividing by λ and taking the infimum over couplings yields $K \exp\{\lambda(M-m)\} e$. Letting m and M approach the continuation's infimum and supremum proves (59). Translation invariance follows from the oscillation rather than the absolute level. The expectation limit follows from boundedness and expansion of the exponential integral; its displayed coefficient tends to K.

If $f(Y) \geq f(\widehat{Y})$ almost surely, every positive-part integral is weakly larger and so is its minimized variational expression. Exponentiation and logarithm preserve the same order for the entropic evaluation. This gives the signed lower endpoint zero. If $|\widehat{Q} - c - \beta \rho_{\widehat{P}}(f)| \leq \xi$, subtracting the numerical approximation places the action-value discrepancy in $[-\xi, \beta\kappa e + \xi]$ under that ordering, or in $[-\beta\kappa e - \xi, \beta\kappa e + \xi]$ without it. These are bounds for the original risk primitive; no statement is made about replacing one primitive by another.

15.4. *Nested objectives and the dated resource account*

Nested one-step evaluations and a static terminal evaluation differ even without decisions. With four equiprobable terminal losses $0, 0, 0, 4$, group the first two and last two under two equiprobable intermediate states. At risk level one half, the intermediate evaluations are zero and four, and the nested value is four. Static average value at risk of the four terminal outcomes is two. Consequently a static

objective needs its own state augmentation or a different analysis; applying the preceding nested recursion without that step changes the objective.

For an originally nested criterion, the preceding induction propagates a state-independent continuation allowance a as $\beta_t a$. The local Wasserstein-to-risk coefficient κ_t enters only the current discrepancy. Substituting the symmetric signed interval into (57) gives (60); monotone-coupling information replaces the factor two on that transition component by one. This establishes the budget used for dated allocation. It does not bound the cost of obtaining K_{t+1} or the number of labels, innovations or feasible-action searches. In particular, a growth of the verified continuation modulus with horizon is retained in κ_t and cannot be removed by this algebra.

For a fixed-opponent best-response problem the identical induction compares a player's implemented complete strategy with the original best-response optimum. Applying it separately in every declared subgame gives the corresponding unilateral-deviation certificate. Each temporal self likewise retains its own original criterion and fixed future strategies. Monotone but non-cash-invariant aggregators do not satisfy the constant-shift step above and must instead use their certified absolute-error recursion. This establishes the advertised scope without conflating expectation, nested risk, static risk and strategic evaluation.

16. PROOFS FOR REFERENCE-CENTERED RISK QUALIFICATION

16.1. *Centering and the original Bellman optimum*

For any bounded z, substituting $f_{t+1} + z$ in (61) and subtracting the finite number $\rho(f_{t+1})$ proves (63). Each dual objective at f_{t+1} is at most its supremum, giving $k^f \geq 0$. Approximating that supremum gives $\inf_q k^f(q) = 0$ without an attainment assumption. Since each q is a probability, adding a constant b to z adds b to $\mathcal{E}^f(z)$. If $z \leq w$, every dual objective at z is no larger than its counterpart at w, proving monotonicity. The inequalities $w - \|z - w\|_\infty \leq z \leq w + \|z - w\|_\infty$ prove nonexpansiveness.

PROOF OF THEOREM 24: Let $d_t^\pi = V_t^\pi - f_t$ and $d_t^* = V_t^* - f_t$. The original recursions and (63) give the exact identities

$$d_t^\pi(x) = r_t(x, \pi_t(x)) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(d_{t+1}^\pi),$$
$$d_t^*(x) = \inf_{a \in A_t(x)} \{r_t(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(d_{t+1}^*)\}. \tag{6}$$

Both have terminal value $g - f_T$. Suppose $d_{t+1}^\pi \le U_{t+1}$. Monotonicity, nonnegative β_t, the upper residual inequality and (64) imply $d_t^\pi \le U_t$. Similarly $L_{t+1} \le d_{t+1}^*$ implies $L_t \le d_t^*$. Backward induction establishes both statements. Feasibility of the supplied complete policy implies $V_t^* \le V_t^\pi$, giving (65). The argument takes place at each fixed parameter vector; applying it to every vector satisfying the premises proves the class statement without assembling incompatible continuations.

Exact equality recursions coincide with (6), proving the exactness assertion by induction. For the scalar comparison, set $a_t = \ell_t + b_t^-$ and $b_t = u_t + \gamma_t + b_t^{\pi,+}$ from the signed assumptions. Every original action has residual at least a_t, since its numerical action value is at least the numerical Bellman infimum. The implemented action has residual at most b_t. Define $\underline{h}_T = d^-$, $\overline{h}_T = d^+$, and

$$\underline{h}_t = a_t + \beta_t \underline{h}_{t+1}, \qquad \overline{h}_t = b_t + \beta_t \overline{h}_{t+1}.$$

Since $\mathcal{E}^f(b) = b$ for constants, these are feasible lower and upper certificates. Exact backward evaluation is order preserving, so $L_t \ge \underline{h}_t$ and $U_t \le \overline{h}_t$ for exact residuals and terminal discrepancies. Their date-zero difference is at most (57).

For two policy supersolutions, their pointwise minimum remains a supersolution: the operator at the minimum of their successor bounds is at most its value at either bound. The corresponding pointwise maximum of two optimum subsolutions remains a subsolution by the reverse argument. Finite combinations therefore preserve an already valid scalar guarantee. No claim about improved bounds from an uncontrolled numerical optimizer is needed. *Q.E.D.*

16.2. *Risk occupations and certificate orientation*

PROOF OF PROPOSITION 25: For fixed π, (6) is the Bellman equation for a finite-horizon maximizer selecting a risk index j after observing the current state. Its

reward is $r_{ts} - \beta_t k_{tsj}$, transition vector is q_{tsj}, and terminal reward is $g - f_T$. Finite maxima have measurable maximizers. Backward induction establishes a Markov maximizing choice at each date and state.

Any nonanticipating choice rule, including a history-dependent one, induces occupation masses satisfying (66). Conversely, for a feasible flow, choose j at (t,s) with probabilities y_{tsj}/m_{ts} when $m_{ts} > 0$; at zero-mass states choose an arbitrary feasible index. Induction on t recovers the declared masses. The corresponding expected sum of rewards is exactly the objective in (67). Thus the flow maximum equals the centered fixed-policy value. This representation is under the adversarial risk kernels and must not be called the physical occupation law.

Multiply each dual inequality by $B_t y_{tsj}$ and sum. The flow equalities cancel intermediate potential terms, leaving the primal objective at most $\mu^\mathsf{T} U_0$. The argument allows a zero discount: it uses $B_{t+1} = B_t \beta_t$ but never divides by either factor. Feasible primal and dual witnesses with equal objectives certify the maximum. For an optimum subsolution, a separate feasible risk witness for every original action supplies a lower bound on that action's supremum; a minimization over a restricted sampled action set does not supply the required all-action inequality. *Q.E.D.*

16.3. *Kernel exclusion and its sharp boundary*

For any $z \in \mathcal{D}$, the difference between the affine pieces indexed by q and q_0 is

$$(q - q_0)^\mathsf{T} z - k^f(q) + k^f(q_0).$$

A strict negative upper bound excludes q uniformly. This proves (68). Maximizing coordinate by coordinate gives the box formula. For a span ball, the coefficients sum to zero; adding a constant to z changes nothing. Normalize $\min_i z_i = 0$ and assign s to the positive coefficients and zero to the negative ones. The maximum is $s \sum_i (q_i - q_{0i})_+ = s\|q - q_0\|_1/2$.

As a sharp example, take two equally likely outcomes and average value at risk at level $1/2$. Then $\rho(v) = \max(v_1, v_2)$ and the dual vertices are e_1, e_2. At $f = (0,1)$ their centered penalties are $1, 0$. For $z = (2,0)$, the centered value is 1. Keeping only the maximizing kernel e_2 from the reference gives 0, which is invalid as an upper

bound; dropping penalties gives 2, which is conservative but not exact. On the span ball of radius $1/2$, the first kernel is safely excluded. At radius 1, $z = (1, 0)$ makes the two pieces equal, so the strict exclusion must retain the boundary. This example proves both the necessity of allowing the risk kernel to change and sharpness of the strict threshold convention.

16.4. *Entropic duality without changing the objective*

Let p be a probability measure, f, z bounded, and $\lambda > 0$. Boundedness makes $Z_f = \int e^{\lambda f} dp$ strictly positive and finite, and makes p_f equivalent to p on its support. For $q \ll p$ with finite relative entropy,

$$\lambda^{-1}\mathrm{KL}(q\|p_f) = \lambda^{-1}\mathrm{KL}(q\|p) - \int f\, dq + \lambda^{-1}\log Z_f.$$

If q places mass outside the support of p, its penalty is infinite. For any bounded v, define $p_v = e^{\lambda v} p / Z_v$. Nonnegativity of $\mathrm{KL}(q\|p_v)$ gives

$$\int v\, dq - \lambda^{-1}\mathrm{KL}(q\|p) \le \lambda^{-1}\log Z_v,$$

with equality at $q = p_v$. This proves the variational representation, its centered penalty, and (69). It also shows exactly why using the fixed tilt p_f as an expectation is not generally exact: the centered evaluation remains entropic under p_f, rather than linear. For $\lambda \downarrow 0$, bounded convergence gives the expectation limit. Exponential and logarithmic evaluations in a numerical certificate require outward arithmetic; the rational average-value-at-risk experiments are not evidence of exact floating-point entropic computation.

16.5. *Occupation-sensitive precision and a strict improvement*

For fixed penalties and transitions, write the flow objective as $A(y)$ and the additional nonnegative residual contribution as $E(y) = \sum_{t,s} B_t e_{ts} m_{ts}$. If $H = \max_y A(y)$ and $H_e = \max_y \{A(y) + E(y)\}$, then

$$0 \le H_e - H \le \max_y E(y).$$

The upper inequality follows from $A(y) \leq H$ for every feasible y; the lower follows from $E(y) \geq 0$. Since each occupation layer has total mass one, $\max_y E(y) \leq \sum_t B_t \max_s e_{ts}$. This establishes (70). The argument assumes the penalties are held fixed while residual uncertainty is enlarged. Altering the underlying stored continuation changes the penalties and requires requalification.

For a strict horizon example, let two states be absorbing under both actions, let action zero cost zero and action one cost one in both states, let $g = 0$, and let all discounts equal one. Every normalized monetary risk evaluation of these deterministic transitions agrees with evaluation at the absorbing state. Put $f_t(s) = (T - t)s$ for $s \in \{0, 1\}$ and choose action zero everywhere. Each date's implemented and minimum residuals are $(0, -1)$; the signed scalar account charges one at each date, yielding T. The exact state-dependent lower and upper recursions both equal $-(T - t)s$, giving zero policy loss at every state. There are two feasible actions and every action is checked. No state-dependent bias was cancelled without propagation: its entire state dependence was carried through the original transition at every date. This is consistent with, and sharpens rather than contradicts, the statewise-bias warning in Section 15.

Finally, the finite backward recursion evaluates each stored transition support once for each retained risk piece at each date and state-action pair. Summing these support sizes yields the stated arithmetic count. A direct descendant recursion repeatedly evaluates the same subproblem; removing that repetition does not establish a polynomial bound in the number of continuous states, risk pieces, acquisition cells or rational bit length. The comparison is a representation-conditioned resource statement, and the full construction and support-qualification work remains separately charged.

REFERENCES

HOEFFDING, WASSILY (1963): "Probability Inequalities for Sums of Bounded Random Variables," *Journal of the American Statistical Association*, 58 (301), 13–30. [11]

JIANG, NAN AND LIHONG LI (2016): "Doubly Robust Off-policy Value Evaluation for Reinforcement Learning," in *Proceedings of the 33rd International Conference on Machine Learning*, vol. 48 of *Proceedings of Machine Learning Research*, 652–661. [2]

MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): "Empirical Bernstein Bounds and Sample Variance Penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [10]